

# SOLUTION DESIGN DOCUMENT

## Mental Health Risk Screening & Support Chatbot

| Field            | Details                                 |
|------------------|-----------------------------------------|
| Student          | Vamsi Krishna Yarragunta                |
| Submission Type  | Individual Class Project                |
| Document Version | v1.2                                    |
| Date             | Aug 28,2026                             |
| Status           | Completed – Dashboard, ML & RAG Chatbot |

### 1. Executive Summary

The Mental Health Risk Screening & Support Chatbot is an AI-powered application designed to analyze workplace mental health data and provide users with meaningful insights and supportive guidance. The project uses the OSMI Mental Health in Tech Survey dataset, containing 1,259 respondents, with the data stored and managed using a MySQL database. The project uses Pandas in Jupyter Notebook for data analysis and identifying important patterns in the survey data, while Plotly and Cufflinks are used to create interactive visualizations. Logistic Regression and Random Forest models are developed to predict treatment-seeking behavior, with ROC-AUC scores of 0.8647 and 0.8764, respectively.

The trained model is integrated into a Streamlit Risk Assessment module, where users can receive an estimated treatment-seeking probability based on their inputs. The application also includes a RAG-based mental health support chatbot using ChromaDB and Sentence Transformers to retrieve relevant information from a curated knowledge base, while Groq LLM generates conversational responses. Overall, the project combines data analysis, visualization, machine learning, RAG, and generative AI into a single interactive platform for workplace mental health insights and general support.

### 2. Purpose of This Document

This document describes the project purpose, requirements, scope, architecture, data and ML pipeline, chatbot design, technology stack, timeline, functional and technical requirements, assumptions, risks, limitations and conclusion.

### 3. Project Background

Workplace mental health can influence employee well-being, work performance and treatment-seeking behavior. Historical survey data can reveal relationships between treatment seeking and factors such as family history, work interference, workplace benefits and awareness of care options.

The project converts these historical patterns into an interactive analytics and screening application while providing a separate RAG-grounded assistant for general support and wellness information.

## 4. Version History

| Version | Date       | Notes                                                                          |
|---------|------------|--------------------------------------------------------------------------------|
| v1.0    | July, 2026 | Data loading, MySQL setup, cleaning, EDA and survey analysis                   |
| v1.1    | Aug, 2026  | Feature preparation, Logistic Regression and Random Forest training/evaluation |
| v1.2    | Aug, 2026  | Streamlit dashboard, visualizations and risk assessment                        |
| v1.3    | Aug, 2026  | ChromaDB RAG, knowledge base, Groq chatbot and integration                     |

## 5. Project Team

| Name                     | Role    | Responsibilities                                                                 |
|--------------------------|---------|----------------------------------------------------------------------------------|
| Vamsi Krishna Yarragunta | Student | Design, development, analysis, ML, dashboard, chatbot, testing and documentation |

## 6. Business Requirements

| # | Requirement              | Detail                                                                      |
|---|--------------------------|-----------------------------------------------------------------------------|
| 1 | Collect and prepare data | Load, clean and prepare the historical OSMI survey.                         |
| 2 | Store project data       | Store prepared survey data in MySQL.                                        |
| 3 | Perform data analysis    | Analyze demographic, workplace and treatment-seeking patterns.              |
| 4 | Create visual dashboards | Provide interactive KPIs, charts, filters and analytical views.             |
| 5 | Build ML models          | Estimate treatment-seeking probability.                                     |
| 6 | Provide risk assessment  | Allow users to enter survey information and receive a statistical estimate. |
| 7 | Add RAG chatbot          | Retrieve relevant knowledge and generate grounded support responses.        |

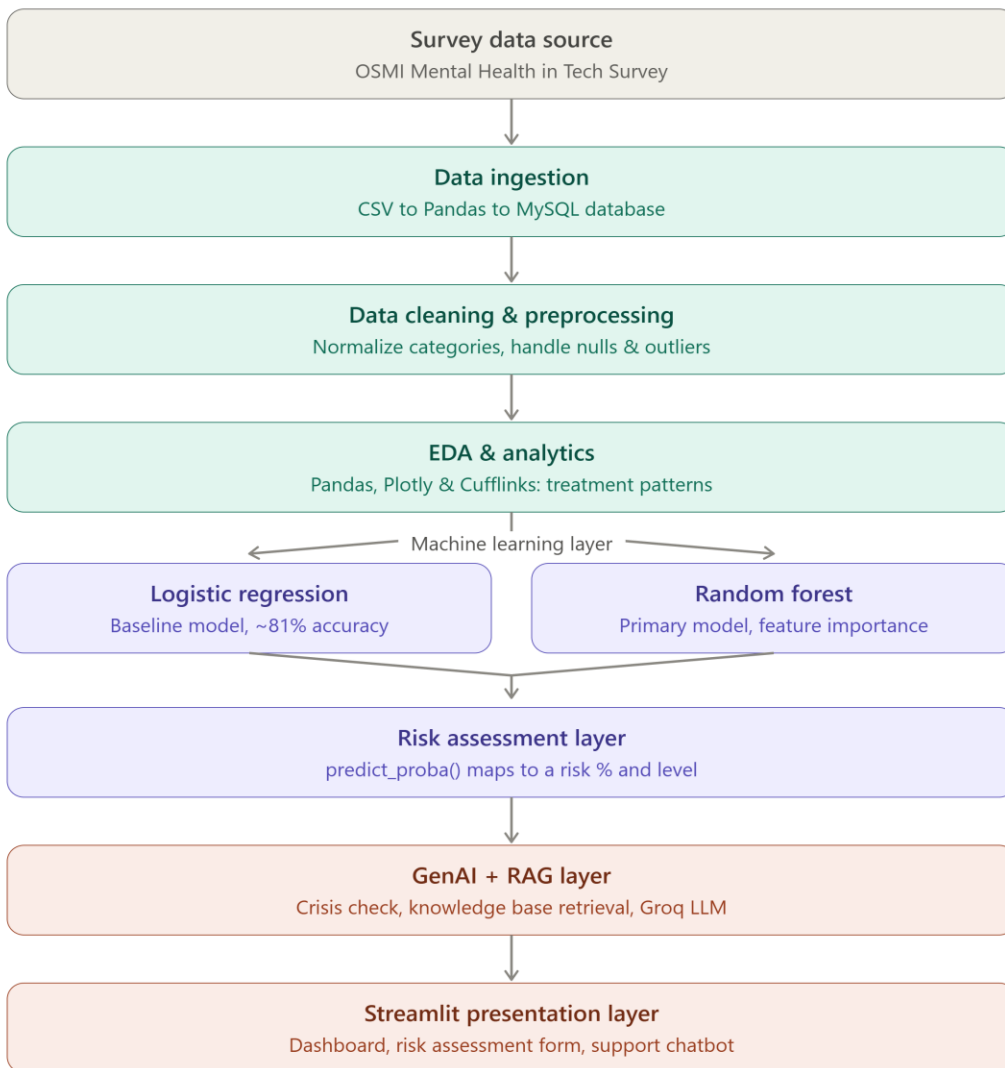
## 7. Scope

### In Scope

- Load and clean the OSMI Mental Health in Tech Survey.
- Store survey data in MySQL.
- Use pandas in Jupyter Notebook for EDA and insight generation.
- Create interactive visualizations using Plotly and Plotly Cufflinks.

- Analyze treatment-seeking and workplace mental-health patterns.
- Train and compare Logistic Regression and Random Forest.
- Evaluate models using ROC-AUC.
- Build a Streamlit dashboard with KPIs, filters and charts.
- Provide a separate treatment-seeking risk assessment.
- Build a lightweight RAG pipeline using Sentence Transformers, ChromaDB and a curated knowledge base.
- Use the Groq API to generate natural-language support responses.

## 8. Architecture



## 9. Success Criteria

- OSMI survey data is loaded, cleaned and stored in MySQL.
- Pandas analysis and Plotly/Cufflinks visualizations reveal meaningful patterns.
- Logistic Regression and Random Forest are trained and compared.
- ROC-AUC: Logistic Regression 0.8647; Random Forest 0.8764.

- Streamlit provides interactive dashboard and risk assessment functionality.
- The chatbot retrieves relevant knowledge-base content before generating responses.
- The chatbot remains separate from the ML risk assessment and does not calculate a new risk score.

## 10. Solution Overview

| Component       | Description                                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Data            | OSMI Mental Health in Tech Survey with 1,259 respondents, stored in MySQL.                                                            |
| Analytics       | Pandas EDA covering demographics, treatment seeking, family history, work interference, benefits, care options and workplace support. |
| Visualization   | Interactive Plotly/Cufflinks charts, KPIs and filters.                                                                                |
| ML              | Logistic Regression and Random Forest for treatment-seeking probability.                                                              |
| Risk Assessment | Streamlit form using encoded inputs and the trained Random Forest probability output.                                                 |
| RAG Chatbot     | Semantic retrieval using Sentence Transformers and ChromaDB followed by Groq LLM generation.                                          |

## 11. How the Pipeline Works

1. The historical OSMI survey is consumed and prepared.
2. Cleaned survey data is stored in MySQL.
3. Pandas is used in Jupyter Notebook for EDA and pattern identification.
4. Plotly/Cufflinks visualizations communicate findings.
5. Categorical variables are encoded and model features are prepared.
6. Logistic Regression and Random Forest are trained.
7. Models are evaluated using ROC-AUC.
8. The trained model and label encoders are loaded by Streamlit.
9. The dashboard retrieves survey data from MySQL and displays KPIs, charts and filters.
10. The risk-assessment form encodes user inputs and calculates treatment-seeking probability.
11. For a non-crisis chatbot question, the question is embedded with all-MiniLM-L6-v2.
12. ChromaDB retrieves the most relevant knowledge-base documents.
13. Retrieved context is added to the chatbot prompt.
14. The Groq-hosted LLM generates the final grounded response.
15. The response is displayed in the Streamlit Support Chatbot.

## 12. Technology Stack

| Technology / Library      | Use                                                              |
|---------------------------|------------------------------------------------------------------|
| Python                    | Core programming and application logic                           |
| Jupyter Notebook          | EDA and ML development                                           |
| pandas / NumPy            | Data processing and numerical computation                        |
| MySQL                     | Structured survey-data storage                                   |
| SQLAlchemy / PyMySQL      | Database connection and retrieval                                |
| Plotly / Plotly Cufflinks | Interactive visualizations                                       |
| scikit-learn              | Preprocessing, Logistic Regression, Random Forest and evaluation |
| pickle                    | Save/load model and encoders                                     |
| Streamlit                 | Dashboard, navigation, assessment and chatbot UI                 |
| Sentence Transformers     | Semantic embeddings                                              |
| all-MiniLM-L6-v2          | Embedding model for retrieval                                    |
| ChromaDB                  | Vector store and similarity retrieval                            |
| Groq API                  | LLM response generation                                          |
| python-dotenv             | Environment-variable credential loading                          |

## 13. Implementation Timeline

| Phase | Focus               | Status    | Work                                                                                                                                                          |
|-------|---------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Data & EDA          | Completed | Data loading, MySQL database setup, data cleaning, exploratory data analysis, and mental health survey analysis                                               |
| 2     | Machine Learning    | Completed | Feature engineering, data encoding, model training, Logistic Regression and Random Forest model comparison, model evaluation, and risk assessment logic       |
| 3     | Dashboard & Chatbot | Completed | Streamlit dashboard, interactive visualizations, risk assessment interface, RAG chatbot, ChromaDB retrieval, Groq GenAI responses, and end-to-end integration |

## 14. Functional Requirements

| ID | The System Must Be Able To                                           |
|----|----------------------------------------------------------------------|
| 1  | Read and prepare historical OSMI mental-health survey data.          |
| 2  | Store and retrieve prepared data using MySQL.                        |
| 3  | Perform pandas-based analysis and identify meaningful patterns.      |
| 4  | Produce interactive charts, KPIs, filters and dashboard views.       |
| 5  | Prepare encoded features for ML.                                     |
| 6  | Train and compare Logistic Regression and Random Forest.             |
| 7  | Evaluate models using ROC-AUC.                                       |
| 8  | Provide a treatment-seeking probability through the assessment form. |
| 9  | Retrieve relevant mental-health knowledge using RAG.                 |
| 10 | Answer support questions through a retrieval-grounded chatbot.       |
| 11 | Provide crisis-oriented guidance for detected crisis queries.        |
| 12 | Keep chatbot support separate from the ML risk assessment.           |

## 15. Technical Requirements

| ID | Requirement                                                                                          |
|----|------------------------------------------------------------------------------------------------------|
| 1  | Required Python dependencies must be installed in the supported environment.                         |
| 2  | The data pipeline should handle missing values, duplicates and inconsistent data types when present. |
| 3  | Prepared data should be reusable by analytics, ML and Streamlit.                                     |
| 4  | Features must be transformed consistently for training and inference.                                |
| 5  | Classification models must be evaluated using suitable metrics including ROC-AUC.                    |
| 6  | The Streamlit application must render without runtime errors.                                        |
| 7  | The chatbot must retrieve relevant knowledge before generating responses.                            |
| 8  | The chatbot must not diagnose conditions or prescribe treatment.                                     |
| 9  | API credentials must be supplied through environment variables and not hardcoded.                    |
| 10 | The knowledge base and vector collection must be available at chatbot runtime.                       |

## 16. Requirement Mapping

| ID | Requirement              | What Is Built                                         | Status       |
|----|--------------------------|-------------------------------------------------------|--------------|
| 1  | Collect and prepare data | OSMI survey loading, cleaning and preparation         | COMPLETE     |
| 2  | Store data               | MySQL survey_data table                               | COMPLETE     |
| 3  | Perform analysis         | Pandas EDA and workplace mental-health analysis       | COMPLETE     |
| 4  | Create visual dashboard  | Streamlit KPIs, filters, Plotly charts and navigation | COMPLETE     |
| 5  | Build ML models          | Logistic Regression and Random Forest                 | COMPLETE     |
| 6  | Evaluate models          | ROC-AUC 0.8647 and 0.8764 respectively                | COMPLETE     |
| 7  | Risk assessment          | Streamlit form with model-based probability           | COMPLETE     |
| 8  | Add RAG chatbot          | Embeddings + ChromaDB retrieval + Groq generation     | COMPLETE     |
| 9  | Clinical diagnosis       | No diagnosis/treatment functionality                  | OUT OF SCOPE |
| 10 | Real-time health data    | Historical OSMI survey; no live clinical stream       | OUT OF SCOPE |

## 17. Assumptions

- The OSMI survey is representative enough to demonstrate workplace mental-health patterns for this academic project.
- The project is a screening and decision-support prototype, not a clinical diagnostic system.
- Required Python, Jupyter, MySQL, Streamlit and libraries are available.
- Historical survey data is sufficient for demonstrating descriptive and predictive analytics.
- The RAG knowledge base contains the approved general mental-health guidance used by the chatbot.
- The Groq API is available when its API key is configured.
- The model output is interpreted as a statistical treatment-seeking estimate, not a medical diagnosis.

## 18. Risks

| Risk                                               | Likelihood | Impact | Mitigation                                                                            |
|----------------------------------------------------|------------|--------|---------------------------------------------------------------------------------------|
| Historical data may not represent all populations. | Medium     | High   | Document dataset limitations and avoid clinical conclusions.                          |
| ML model may overfit historical patterns.          | Medium     | High   | Use train/test separation and appropriate evaluation.                                 |
| Probability may be misunderstood as diagnosis.     | Medium     | High   | Clearly label it as a statistical estimate.                                           |
| LLM may generate unsupported information           | Medium     | High   | Use RAG retrieval and constrained system instructions.                                |
| Crisis wording may vary.                           | Medium     | High   | Use multiple crisis indicators and direct users to emergency resources when detected. |
| API key exposure.                                  | Medium     | High   | Use environment variables.                                                            |
| Dependency/database configuration errors.          | Medium     | Medium | Maintain dependencies and test the application.                                       |

## 19. System Limitations

- The implementation uses historical survey data and is not a real-time health surveillance system.
- The dataset may not generalize to every population or organization.
- The risk assessment estimates treatment-seeking probability and does not diagnose a mental-health condition.
- The chatbot is for general information, wellness guidance and resources, not a replacement for a professional.
- The RAG implementation uses a lightweight local/in-memory ChromaDB collection.
- The chatbot knowledge is limited to the curated knowledge base.
- LLM response quality depends on retrieved context and the Groq service.
- The application is an academic prototype and is not connected to clinical or employer systems.

## 20. Conclusion

The Mental Health Risk Screening & Support Chatbot combines historical workplace mental-health data analysis, interactive visualization, machine learning and retrieval-grounded GenAI in one Streamlit application. MySQL stores the survey data; pandas and Plotly/Cufflinks support analysis and visualization; Logistic Regression and Random Forest provide treatment-seeking probability estimates.

Random Forest achieved a ROC-AUC of 0.8764, while Logistic Regression achieved 0.8647. The RAG chatbot uses all-MiniLM-L6-v2 embeddings and ChromaDB to retrieve relevant information from a curated knowledge base before the Groq LLM generates a response. The chatbot is intentionally separated from the risk assessment so it provides supportive information rather than performing another risk score.